CSA0919 PROGRAMMING IN JAVA
HARD LEVEL PROGRAMS(1-15)

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1

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int days = sc.nextInt();

        if(days < 0){
            System.out.println("Invalid Input");
            return;
        }

        int years = days / 365;
        days %= 365;
        int weeks = days / 7;
        int rem = days % 7;

        System.out.println("No. of years: " + years);
        System.out.println("No. of weeks: " + weeks);
        System.out.println("No. of days: " + rem);

        int d = sc.nextInt();
        int m = sc.nextInt();
        int y = sc.nextInt();

        Calendar cal = Calendar.getInstance();
        cal.set(y, m-1, d);

        String[] arr = {"Sunday","Monday","Tuesday","Wednesday","Thursday","Friday","Saturday"};
        System.out.println(arr[cal.get(Calendar.DAY_OF_WEEK)-1]);
    }
}
```

input

```
756
No. of years: 2
No. of weeks: 3
No. of days: 5
 31 8 2019
Saturday
```

2

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int total = sc.nextInt();
        int staff = sc.nextInt();

        if(total < 0 || staff < 0 || staff > total){
            System.out.println("Invalid Input");
            return;
        }

        int nonTeaching = staff / 3;
        int students = total - (staff + nonTeaching);

        System.out.println("Student Users: " + students);
    }
}
```

input

```
856
126
Student Users: 688
```

3

```java
        int k = sc.nextInt();

        if(n <= 0 || k <= 0){
            System.out.println("Invalid Input");
            return;
        }

        ArrayList<Integer> list = new ArrayList<>();

        for(int i=1;i<=n;i++){
            if(n%i==0) list.add(i);
        }

        System.out.println("Number of factors = " + list.size());

        if(k <= list.size())
            System.out.println(k + "th factor of " + n + " = " + list.get(k-1));
        else
            System.out.println("No such factor");
    }
}
```

input

```
100
4
Number of factors = 9
4th factor of 100 = 5
```

4

```java
                System.out.println(n+"th Prime number is "+num);

                int next=0;
                int temp=num+1;

                System.out.print(n+" prime numbers after "+num+" are: ");

                while(next<n){
                    if(isPrime(temp)){
                        System.out.print(temp+" ");
                        next++;
                    }
                    temp++;
                }
                break;
            }
        }
        num++;
    }
}
}
```

input

```
3
3th Prime number is 5
3 prime numbers after 5 are: 7 11 13
```

5

```java
        ArrayList<Integer> res = new ArrayList<>();

        for(int i=l;i<=u;i++){
            int s = (int)Math.sqrt(i);

            if(s*s == i){
                int sum=0, temp=i;

                while(temp>0){
                    sum += temp%10;
                    temp/=10;
                }

                if(sum < 10)
                    res.add(i);
            }
        }

        System.out.println(res);
    }
}
```

input

```
1
40
[1, 4, 9, 16, 25, 36]
```

6

```java
        }

        for(int i=0;i<str.length();i++){
            char ch = str.charAt(i);
            String rem = str.substring(0,i) + str.substring(i+1);
            permute(rem, ans+ch, set);
        }
    }

    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.next();

        if(s.contains("-")){
            System.out.println("Invalid Input");
            return;
        }

        permute(s, "", new HashSet<>());
    }
}
```

input

```
143
143
134
413
431
314
341
```

7

```java
import java.util.*;

public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int l = sc.nextInt();
        int u = sc.nextInt();

        if(l > u){
            System.out.println("Invalid Input");
            return;
        }

        for(int i=l;i<=u;i++){
            System.out.print("(" + i + ", " + (i*i) + ") ");
        }
    }
}
```

input

```
45
49
(45, 2025) (46, 2116) (47, 2209) (48, 2304) (49, 2401)
```

8

```java
    }
}

public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        Account a = new Account();

        a.accNo = sc.nextInt();
        a.name = sc.next();
        a.type = sc.next();

        double amt = sc.nextDouble();

        a.deposit(amt);
        a.withdraw(amt);

        a.display();
    }
}
```

input

```
100
Raja
S
8000
Balance: 10000.0
```

9

```java

        if(num < 0){
            System.out.println("Invalid Input");
            return;
        }

        while(true){
            String rev = new StringBuilder(""+num).reverse().toString();
            long sum = num + Long.parseLong(rev);

            System.out.println(num + " + " + rev + " = " + sum);

            if(isPalindrome(""+sum)){
                System.out.println("Palindrome: " + sum);
                break;
            }

            num = sum;
        }
    }
}
```

input

```
7325
7325 + 5237 = 12562
12562 + 26521 = 39083
39083 + 38093 = 77176
77176 + 67177 = 144353
144353 + 353441 = 497794
Palindrome: 497794
```

10

```java
public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int w = sc.nextInt();
        int d = sc.nextInt();

        if(w < 0 || d < 0){
            System.out.println("Invalid Input");
            return;
        }

        Customer c = new Customer();

        new Thread(() -> c.withdraw(w)).start();

        try{ Thread.sleep(1000); }catch(Exception e){}

        new Thread(() -> c.deposit(d)).start();
    }
}
```

input

```
1200
3000
Withdraw success, balance: 8800
Deposited: 3000
```

11

```java
import java.util.*;

public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for(int i=1;i<=n;i++){
            if(i%3==0 && i%5==0)
                System.out.print("FizzBuzz ");
            else if(i%3==0)
                System.out.print("Fizz ");
            else if(i%5==0)
                System.out.print("Buzz ");
            else
                System.out.print(i+" ");
        }
    }
}
```

```
5
1 2 Fizz 4 Buzz
```

12

```java
        int n2 = sc.nextInt();
        int[] b = new int[n2];

        for(int i=0;i<n2;i++)
            b[i] = sc.nextInt();

        HashSet<Integer> set = new HashSet<>();
        ArrayList<Integer> res = new ArrayList<>();

        for(int x : a) set.add(x);

        for(int x : b){
            if(set.contains(x)){
                res.add(x);
                set.remove(x);
            }
        }

        System.out.println(res);
    }
}
```

input

```
4
1 2 3 4
5
2 4 5 6 7
[2, 4]
```

13

```java
import java.util.*;

public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        String s = sc.nextLine().trim();

        if(s.length() == 0){
            System.out.println(0);
            return;
        }

        String[] arr = s.split(" ");
        String last = arr[arr.length - 1];

        System.out.println(last.length());
    }
}
```

```
Hello world
5
```

14

```java

        for(int i=0;i<s.length();i++){
            char ch = s.charAt(i);

            if(!map.containsKey(ch)){
                System.out.println("Invalid Input");
                return;
            }

            int val = map.get(ch);

            if(i < s.length()-1 && val < map.get(s.charAt(i+1))){
                sum -= val;
            } else {
                sum += val;
            }
        }

        System.out.println(sum);
    }
}
```

```
MCMXCI
1991
```

15

```java
            if(Character.isLetter(c))
                count[c - 'a']++;
        }

        for(char c : ransom.toCharArray()){
            if(Character.isLetter(c)){
                count[c - 'a']--;

                if(count[c - 'a'] < 0){
                    System.out.println(false);
                    return;
                }
            } else {
                System.out.println(false);
                return;
            }
        }

        System.out.println(true);
    }
}
```

```
 a a
true
```